Automatic Category Suggestion for Wikimedia Commons

Megathon submission · Oct 10, 2026

1. Problem statement and our understanding

Commons gets tens of thousands of uploads a day, many under-categorized. Given a new image and its metadata, the system returns a ranked list of categories for a volunteer to accept or reject. We treat it as **extreme multi-label ranking**:

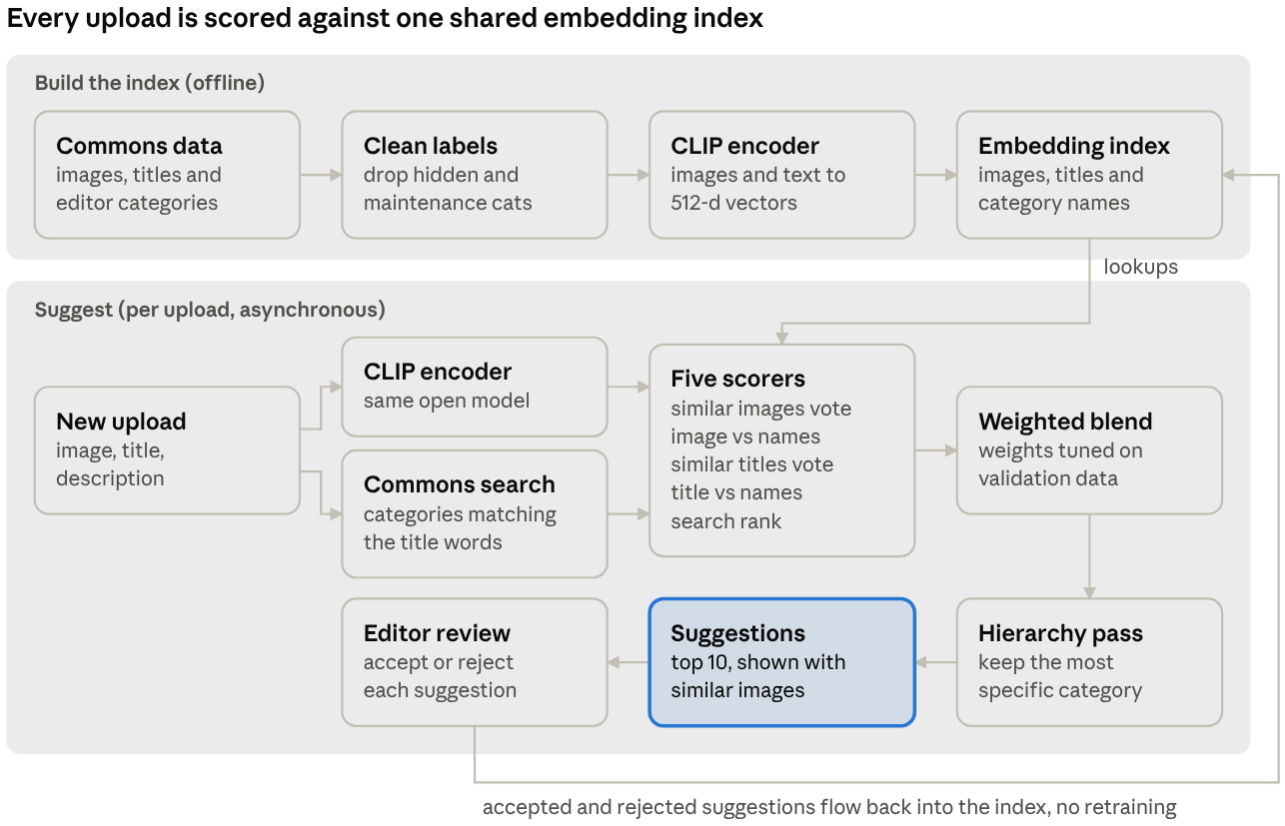
- **Long-tailed label space:** millions of categories; in our 5,966-image sample, 5,021 of 6,314 occur once, so categories are retrieved and ranked, not classified.
- **Incomplete labels:** a missing label is unknown, not wrong.
- **Mixed semantics:** objects show in pixels; events and abstract ideas need the title, description and date.
- **Constant change:** new categories must work without retraining, cheaply and explainably.

2. Idea and abstract

One open model, CLIP ViT-B/32 (LAION-2B weights), maps images, titles, descriptions and category names into one 512-d space. Five scorers rank candidates: **similar images vote**, **image vs category name**, **similar titles vote**, **title vs category name**, and **Commons search** on the title words, which reaches places and events never seen in training. A blend tuned on validation data merges them. Nothing is trained per category, so new images and categories work once embedded, and the similar images behind a suggestion explain it.

3. Approach, pipeline and architecture

- **Offline:** the MediaWiki API supplies images, text and editor categories; hidden and maintenance categories are dropped; CLIP embeds everything into an approximate nearest-neighbour index (FAISS with product quantization holds 100 million images in a few GB, with millisecond lookups).
- **Per upload:** asynchronous, so uploads are never blocked; roughly one image per second at peak fits on one GPU. A hierarchy pass drops a parent when a more specific child is suggested.
- **Learning from new data:** accepted images join the index at once; rejected suggestions become hard negatives for an optional supervised head on frequent categories.



system architecture · offline index build and per-upload suggestion

4. Testing methodology

- **Leak-free splits:** split by upload time (70/10/20), near-duplicates removed, unseen uploaders reported apart, weights tuned on validation only.
- **Metrics:** P@k and R@k (k = 1, 5, 10) per image, macro mAP per category, and recall on labels beyond our 43 seed categories.
- **Breakdowns:** category type, image degradation (blur, JPEG, downscale), region and language, latency and memory, plus a hand review of "wrong" suggestions.

Preliminary results (1,191 newest uploads as test, 4,177 older ones for training): text adds 23 points of top-1 accuracy, and search finds 6× more non-seed labels at almost no top-1 cost. Unseen uploaders score the same, so this is not leakage.

Method	P@1	R@5	R@10	R@10, non-seed labels
Image only	0.427	0.366	0.406	0.026
+ title and description	0.662	0.496	0.513	0.030
+ Commons search (all five)	0.647	0.550	0.576	0.188